

CHAPTER 11

A forefront insight into the integration of AI and blockchain technologies

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1. The rise of blockchain technology

Blockchain, which is expected to store 10% of the world's GDP by 2025, is no more just a concept. If 2016 was the era of blockchain prototypes, 2017 saw the mass adoption of blockchain globally across all business and government fields (Rahul, n.d.). A viable case that any tech expert knows about regarding blockchain technology is the digital Bitcoin cryptocurrency.

Today, it is a multibillion-dollar industry where businesses that range from technology giants to manufacturers spend heavily to develop their environments (Brodersen et al., 2019).

While blockchains were initially viewed as a financial innovation, i.e., introducing cryptocurrencies such as Bitcoin, they have proven to be far more useful in other industrial contexts as well as institutional, legal, and government cases.

1.1 Why is blockchain so popular?

Inherently, blockchain has three main features (Rahul, n.d., Fig. 11.1) that make it a rosy option for private and public actors.

- **Decentralized control:** The blockchain runs on a mutual control system, providing various players with a common mechanism to agree on a common cause. There are no exclusive rights given to any entity in a blockchain network, which allows for the smooth creation of consortia or decentralized networks for various transactional services. It is highly prominent wherever the most competent practices must be accessible to

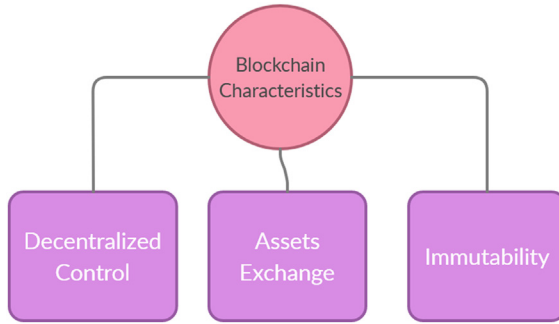


Figure 11.1 The three main features of blockchain.

all sectors to strengthen the system and promote better enforcement of and compliance by all key players.

- **Asset exchange:** This could be the most extensive feature of blockchain because all transactions are registered, and no person has complete authority over what is going on in the network, allowing the transparent interchange of value objects, i.e., value units such as digital money, energy unit data, storage unit data, etc. A trusted network for essential data sharing with blockchains becomes easier to manage. This will dramatically revolutionize how blockchain technology transactions are carried out compared to today's conventional system. More reliability, protection, and straightforward monitoring will ensure that transactions are feasible and that ethical policies are in place.
- **Immutability:** Once registered, the transaction is irreversible and cannot be removed from the network. It is like carving the information in stone. A robust digital audit trace helps everyone in the network identify who carries out activities in that network and provides smart contract management data in which contractual obligations become easier for all parties involved to verify. Ideally, this means the total exclusion of illegal activities and verification of all transactions on such networks. No person nor group would be able to manipulate the whole network in a flawed transaction, and whatever the size of the network, all previous operations will be subject to public monitoring and review by all concerned participants and stakeholders.

These three key features lay the groundwork for a new technology infrastructure planned to trigger dramatic changes in the global economy on a scale seen when the internet achieved renown 2 decades ago.

2. Artificial intelligence

Artificial intelligence (AI) has become a part of everyday life today. For instance, taking an Uber to the airport is not at all uncommon for someone, enabling them to use the time in the car to catch up on email and do some online shopping, appearing at their destination after a productive, smooth experience.

Less than a decade ago, it would have been unusual to contemplate an experience like this. This sort of experience is possible due to AI breakthrough technologies nowadays. AI empowers people to engage with one another, locate an optimal route through traffic, and discover the correct route. It prioritizes email and screens bogus messages while providing consumers with product and service ratings. Finally, it delivers a tailored music profile to the rider. Even more than that, autonomous vehicles aided by AI are about to be licensed for wide-scale implementation on public roads, which means that vehicles can be driverless.

Intelligent machines form a broad ecosystem of technologies, including advanced process automation, predictive analytics, mechanical automation, and robotics. The proliferation of AI has accelerated in the personal and business-to-consumer (B2C) settings in recent years, and it will soon have a substantial impact on the corporate sector.

AI will interrupt company operations by helping companies:

- make educated choices
- use current data better
- create new perspectives from vast volumes of human data
- improve learning
- automate routine, transactional, judgmental tasks
- minimize human physical presence
- develop new programs and services

Two recent events provide evidence of how far AI capabilities have advanced. According to recent reports, an institution of higher learning created “Jill Watson,” an AI-based teaching assistant that answered emails (a system that goes by the name of “Google Duplex”) ([Korn, 2016](#)) and an application that booked a restaurant table over the phone ([Hyken, 2018](#)). In both cases, individuals interacting with the systems failed to recognize that they were talking to a robot.

Despite various conditions and feelings that must be considered, AI is utilized to handle routine scheduling, rule-based operations, and steadily more sophisticated duties, such as stock market investment management. More than fifty percent of hedge funds utilize AI to make investment choices, and over a quarter use it to perform deals without human input (Mcdowell, 2020). Overall, today's AI appears to emulate the human brain to perform basic tasks and make conventional digitization and automation more futuristic, allowing the human workforce to be more productive, efficient, and innovative.

For example, AI enhances the human workforce by capturing and using existing information in automated algorithms and generating new perceptions through machine learning. Consequently, AI can play an essential role in maintaining and driving productivity and development in places where there is a shrinking workforce.

AI can disrupt any part of today's business model (administration, customer engagement, logistics, etc.) and open the door to new services and products. For instance, it might allow companies to do the following (Elser et al., 2018):

- perform highly focused, proactive sales with AI-based customer-estimating and demand sensing, which could deliver products before ordering;
- minimize price leakage by improving product quality and innovation cycles, using data-driven R&D;
- improve consumer awareness and better define potential applications or markets for current and new goods;
- improve production throughput by ensuring correct first-time performance, targeting reaction output, streamlining changeover, and advancing maintenance that eliminates unplanned downtime;
- minimize consumer churn rates by 24/7 customer support, with higher data quality levels;
- Use on-site, natural language-enabled, virtual sales agents that are available to communicate often with customers on issues and remind them of the latest offers.

AI can dramatically impact individual job profiles and the overall workforce by allowing new modern productivity levels and adequate effectiveness in all business functions and operations. AI would perform several transactional, necessary tasks, allowing people to concentrate on more value-added work requiring nuanced imagination and judgment.

3. The arrival of AI in blockchain

While blockchain technology is a revolutionary step within itself, the incorporation of AI opens an overabundance of never-before-seen opportunities that can be drawn into by private and public sector stakeholders (Rahul, n.d.). AI networks are today constituent of many large-scale technology structures. Around you, in all digital technology, there is a bit of AI. While searching to buy a dress from Amazon, your decisions are being influenced by algorithms that have been developed to accommodate every kind of person.

Integrating these two leads to solutions for problems that have been troubling entrepreneurs for a long time. Blockchain offers a way to share value-embedded data without friction. AI allows routing data into action to generate value without the effort of humans. To preserve immutability in a blockchain network, AI can be used as the governing factor, thus building one of the world's most stable ecosystems for data exchange and transactions.

3.1 Integration of artificial intelligence and the blockchain

Intriguingly, AI and blockchain vary in various ways philosophically (Pinto, 2018):

- In contrast to the open, distributed nature of blockchain, AI has a more centralized infrastructure.
- While many AI technologies are owned and run by centralized vendors, most blockchain players on the market publish open-source code, publicly accessible for anyone to search.
- AI is more of a black box approach, while blockchain is more open in all the processed transactions.
- AI is based on probabilistic formulas, while blockchain in nature is more deterministic.

AI start-ups, including IBM, Apple, Amazon, Alibaba, Facebook, Google, and Intel, acquire customers for themselves rather than other companies (Bass and Brustein, 2020). Data give these organizations an insurmountable advantage in training their AI agents. Since data and capabilities are mutually exclusive, the data and capabilities are entirely inaccessible to the rest of the world.

Because of the centralized artificial intelligence, such as massive surveillance with facial recognition and computer-based technology, people

have room to do things they should not (Feldstein, 2019). At the same time, organizations relinquish control of their data security and hand it over to third parties.

3.2 Merging AI and blockchain

Because it can help overcome several AI challenges, blockchain is relevant here. We deploy numerous AI and machine learning (ML) tools to recognize and authenticate users' blockchain identities.

Experts are currently researching ways in which blockchain can be used to build a decentralized AI marketplace. This enables people to freely share their PII (personally identifiable information), ensuring that it will stay safe and private through decentralization and secure computing provided by the blockchain. In reality, users can easily exchange sensitive information such as financial and health data, and only the intended service provider will be able to decode and decrypt user PII with the user's express consent. We assume that massive data will accumulate and keep large corporations competitive through the use of AI algorithms.

An article on Hackernoon lists some new ventures integrating blockchain and AI to build state-of-the-art solutions (Pinto, 2018). One of the most remarkable is SingularityNET, an AI marketplace where companies can acquire AI capacity globally to increase storage growth. Namahe AI is also a platform designed to enhance supply chain productivity through the combination of AI and blockchain to ensure that processes are tracked seamlessly in real time, helping to catch irregularities and fraud in real-time as well. Numerai, an AI-based hedge fund, supports competitions to create and apply prediction models and solutions for industry enthusiasts.

3.3 Why is there great potential for AI-integrated blockchain?

AI's passion for consuming as much data as possible makes it a blockbuster for all stakeholders and improves blockchain's reputation providing the data. The following ecosystems can be generated by combining the two (Rahul, n.d.).

1. **Improved business data models:** While current businesses powered by AI models depend on the information provided by organizations and their ecosystem partners, open data sharing has its problems like privacy concerns, fraudulent contracts, and misuse of data, thereby prohibiting the implementation of entirely accessible data systems. AI requires practically no barriers to access knowledge from the market environment to

make the most accurate decisions. Blockchain systems will have this frictionless access to information as all stakeholders exchange information, and no single individual owns the network. All key players can easily integrate data sources when AI is driving analytics or insight mining. The way the information is used will be specified in the blockchain directory for the entire transaction, so data holders are unable to be exposed to situations where the integrity and value of their data are lost. When data is continuously supplied from many stakeholders, AI systems can immerse themselves and investigate patterns, attitudes, and other aspects to reveal insights never seen before. More players will allow more opportunities to gain reliable insights. The practices will prepare them to adapt more efficiently to the situations based upon how every player responds by assessing best practices and finding the best solutions for a complex scenario.

2. **Newer insights and discovery:** With open access, data caches are combined efficiently, which will help handle new AI data combinations and uncover new data trends and behavior. It is easier to remove false statements and practices with blockchain, and AI can also generate new trustworthy classifiers and filters for scientific data information. This is especially important for retail and predictive customer involvement scenarios in which more data sets will allow companies to gain more traction in sales. A lot of marketing automation can also be accomplished by the integration of blockchain-centric customer data acquisition. Smart business systems may feed professional and unstructured leaders into comparing data to information gathered from other forms of conversation and interaction, including social media, email campaigns, etc., to arrive at more accurate sales opportunities. This will allow quicker sales cycles, provide more meaningful filters to classify the desired product for end customers, and generate more effective shopping stories. This will lead to higher expectations of satisfaction and rising loyalty. Branding support from ordinary people would reinforce retail brands more than any other way.
3. **Smart forecasts:** AI has done a lot to predict results based on operational ecosystem data. However, such forecasts may often be inaccurate because of faulty device data generation, dishonest data source manipulation, or flawed analytical methods used in AI-driven systems. With blockchain, data generators and analytical methods would easily be authenticated since they must accept existing best practices and contracts. Those criteria will be continuously checked in a decentralized

framework. This helps AI systems operate only on genuine and authentic data sets, leading to accurate predictions. More genuine data will make AI systems work more effectively. These observations will then be used to arrive at accurate decisions through deep learning algorithms more factually, which boosts the estimation of end-user behavior for organizational or governance-related means. As forecasts become wise, it undoubtedly aligns marketing and awareness strategies in the right direction, thus ensuring that companies have substantial returns on these fronts. Given that almost 40% of marketers in a recent Hubspot survey have pointed out that providing return on investment in marketing is a significant challenge, the use of blockchain AI-driven systems for smart predictions is critical under current market conditions.

4. ***Digital intellectual property rights:*** Models have helped AI build exceptional success stories that motivate others. However, more frequently, such model designers refuse to disclose their critical data model information for lack of copyright and security policies. Although copyright policies exist, data models intelligently reveal entirely distinct architectures so even original data models cannot be identified. With the integration of blockchain technology, developers can easily allow others to share their data models without losing exclusive rights and patents. Data and related models can be stored with the owner's cryptographic digital signature, for a tamper-resistant global registry. What makes it nearly impossible for someone to assert authorship of creations using these publicly available data is that the integrated AI tools allow study of these data models' patterns and behaviors and discovery of robust insights. Also, companies and data owners are authorized for an interim duration to assert their rightful privileges and intellectual property rights. This provides an engaging way of monetizing innovative knowledge or content in varied streams for professionals.
5. ***Self-governing organizations:*** The dream scenario of AI is a world in which computers carry out tasks without human action. All that is needed to set up an AI-enabled training and learning environment that is globally validated is a working policy and instructions to be secure, tamper-resistant, and not owned by a single entity. This system is simpler than blockchain, and AI systems can read data and process and execute professional assignments quickly. Smart contracts on decentralized blockchain networks control chances of failure or malfunctioning and adverse operational effects for the entire system. The network nodes store the entire operations system for the states, parameters, behavior,

performance, etc., and enable AI systems to behave in compliance with predetermined intelligent policies. Such an autonomous level of service has never been seen before except in science fiction films from the past. However, this autonomous operating mode is very feasible with AI that enables protected blockchain systems with data. Variations from planned actions occur when the system is needed to regulate nodes supervised by all actors. Secure and transparent control mechanisms ensure that things don't go haywire at any time, and human intervention is kept to a minimum or eliminated in ideal circumstances. Entire bureaus supporting back-office operations can be run as autonomous units with scalable AI-enabled blockchain systems without human resources.

4. The possibilities of AI integrated with blockchain

Notwithstanding all the advantages of new technologies, they are not without drawbacks. Though blockchain suffers from scalability, performance, and safety deficiencies, AI has a share of concerns about data protection, description, and confidentiality. The integration of these two capable technologies will help each other complement and revolutionize the next digital generation. AI will help build a ML framework on the blockchain network through its integration, enhancing scalability, security, efficient governance, and personalization. On the other hand, blockchain technology can contribute to the privacy, clarification, and trust of AI (Dinh and Thai, 2018). Fig. 11.2 shows the resulting features of the integration of AI and blockchain.

4.1 Artificial intelligence for blockchain

A blockchain platform's functioning implies various parameters and compensations between efficiency, decentralization, safety, and many characteristics. Incorporating AI will help make these decisions simpler by optimizing and automating blockchains to boost performance. All data on the blockchain openly allows AI to play a prominent role in maintaining user privacy and confidentiality (Dinh and Thai, 2018).

1. **AI for security and scalability:** Blockchains are considered immutable and virtually impossible to hack. However, the same cannot be said for the applications developed on the framework, as shown in Decentralized Autonomous Organization (DAO) hacking. With the unbelievable advancement of ML, an AI-run blockchain can detect the possibility of an attack and use the proper defense to handle it. AI can

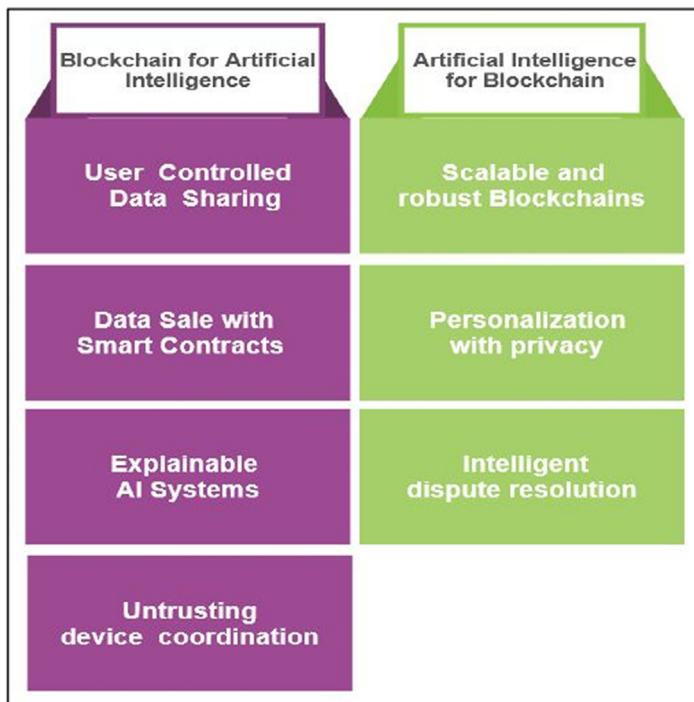


Figure 11.2 AI and blockchain integration.

also be used for the scalability and robustness of the blockchain framework. For instance, the platform may incorporate intelligence to increase the block development rate when transactions spike. This would help to improve the system's efficiency.

2. **AI for privacy:** Today, we collect information about our users through apps to customize content on the site. Enabling blockchain ML information can help to provide privacy while personalizing content. Instead of moving it as conventional centralized systems, the system would pull, view, and display the information. It conducts all local calculations, removing the need to pass data to an external source while protecting user privacy.
3. **AI for smart contracts:** Smart contracts are central to the management of blockchain transactions. However, they are not smart enough to settle conflicts at present. The progress of ML enables AI to solve such complex situations. By incorporating AI into blockchain, users can settle both off-chain and on-chain litigations without third-party

intervention. An AI-backed blockchain platform would execute automated arbitrations in a manipulative and impartial way. The decisions made are guided by data, making them more justified and consistent.

4.2 Blockchain for artificial intelligence

Blockchain can manage coordination mechanisms and decentralized markets for various AI components, including computing power, algorithms, and data. This will promote AI's adoption, creativity, and openness to an unparalleled degree (Dinh and Thai, 2018).

1. **Blockchain for user-controlled data sharing:** Data in the present data-driven economy is regarded as the new gold. Huge data availability is one of the guiding forces of AI systems. However, extracting data from training models is difficult even as a significant player, primarily because of current privacy issues. Blockchain incorporation, with its inherent accountability and transparency, can promote data sharing. Users should also be aware of all data information, such as who downloaded or accessed the data. Blockchain helps to give the consumer more trust in the management of data for sharing. The future effect of such data sharing would be immense.
2. **Blockchain for data sales with smart contracts:** Blockchain integrations with AI will allow data sales using smart contracts. This sets the platform for a safe, private data market without intermediaries. These markets are the foundation for all players, reducing price barriers and promoting creativity.
3. **Blockchain for explainable AI systems:** While ML has succeeded in developing autonomous systems, users are reluctant to implement them. This is because many of the actions taken by these processes cannot be demonstrated to people. An immutable trail must be developed to improve adaptability that tracks the data flow in these complex systems. By monitoring decision-making and data treatment at any stage, blockchain technology allows this. This understanding of a straightforward direction will allow people to grasp the whole process better.
4. **Blockchain for untrustworthy devices:** Coordinating and making decisions together, devices such as mobile phones, internet of things (IoT) devices, or swarm robotics have bright potential for AI in the future. These systems can be used from teamwork robots on tactical missions

to software upgrades for devices such as refrigerators. Integrating such platforms with blockchain would boost safety and defend networks against future hacks.

They still have a long way to go for all the innovations and advances in the blockchain and AI space. Technologic developments in algorithms, protocols, computational resources, and data management are inadequate to solve the challenges of the human environment. However, AI and blockchain's integration provides infinite opportunities that further research and development will realize on the future path.

5. Applications of AI and blockchain

The following are the common AI and blockchain applications (Banafa, 2019). Fig. 11.3 shows a list of AI and blockchain applications.

1. **Smart computing power:** You would need a considerable amount of processing power to run a blockchain on a device with its whole encrypted data. AI helps us to step beyond this and to handle tasks more smartly and productively. An example of the hack algorithms used to mine Bitcoin blocks follows a “brutal force” strategy, which systematically lists all possible solution candidates and tests whether all candidates meet the problem assertion to verify a transaction (Pollock, 2018). Guess a learning-based computer algorithm that can glint its real-time ability if sufficient training information is provided (Pollock, 2018).

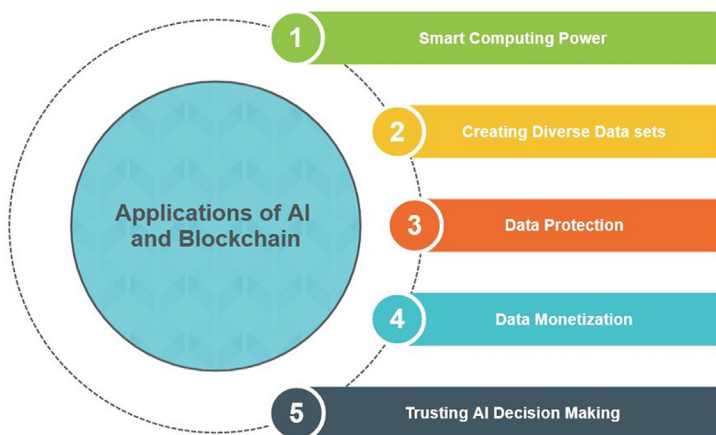


Figure 11.3 Applications of AI and blockchain.

2. **Creating diverse data sets:** Blockchain technology produces decentralized, open networks accessible worldwide in public blockchain networks, unlike AI-based initiatives. However, blockchain network technology is used for decentralization in various industries to supply cryptocurrencies. SingularityNET, for example, is explicitly designed to use blockchain technology to facilitate disseminating algorithms and data, help ensure future AI, and to create decentralized AI. SingularityNET combines AI with blockchain and creates decentralized smarter AI networks that host different data sets. By building an API on the blockchain, an AI intercommunication will be possible. Various algorithms can be developed for data sets (Wolfson, 2018).
3. **Data protection:** AI's success solely depends on the data input. Via data, AI receives valid information and events about the planet. Data feeds AI, which will continue to develop itself through the data. On the other hand, blockchain technology enables secure data storage in a distributed directory. It facilitates the development of entire protected databases to be reviewed by approved parties. By integrating blockchain with AI, a backup mechanism for beneficial and essential personal data is adopted. Financial and health knowledge is too sensitive to be passed to a single entity and its algorithms. The storage of this data on a blockchain accessible by an AI will give us the immense advantage of personalized advice when storing sensitive data securely once we have undergone the proper procedures.
4. **Data monetization:** Monetization of data is another disruptive breakthrough possible by integrating the two technologies. Monetizing collected data is a significant income source for major firms like Google and Facebook (Wolfson, 2018). It reveals that data is being armed against us when others determine how data is sold to profit corporations. Blockchain helps us encrypt our data and use it in the way we think it is appropriate without sacrificing our details and allows us to monetize personal data if we choose. To combat prejudicial algorithms and create a wide variety of data sets in the future, this is necessary to understand (Wolfson, 2018). To learn and improve AI algorithms, AI networks are expected to directly buy data from data markets through their creators and make the whole process even more precise than today without technology firms taking advantage of their users (Wolfson, 2018). Such a data market will also open up AI for small businesses.

For firms that do not produce their data, AI production and feeding are incredibly costly. Using decentralized data markets, they may access costly and privately held data.

5. **Trusting AI decision-making:** As AI algorithms get smarter through learning, data scientists gradually find it increasingly difficult to interpret these programs' implications. This is because AI algorithms can process unbelievably large quantities of variables and data. To ensure that they nevertheless represent the truth, we must continue to audit AI findings. The variables, data, and processes used by AIs for decision-making are stored immutably using blockchain variables. This makes auditing the whole process much more straightforward. All steps are observed using acceptable blockchain programming from data entry to conclusions, and the observer can ensure that these data have not been changed. It builds trust in the AI program's conclusions. It is crucial because using AI apps by firms and individuals will start when they understand how to operate using their knowledge decisions.

The fusion of AI and blockchain technology remains mostly unknown. While there has been a fair amount of wisdom in integrating the two technologies, initiatives dedicated to this innovative alliance are still constrained. These technologies can be combined to use data in ways never before thought possible.

Data are the main component for creating and enhancing AI algorithms, and blockchain means that this data can be audited, any intermediate measures taken by AI to conclude the data can be taken, and the individuals can monetize their data. AI can be unbelievably innovative, but it needs to be planned with the utmost precautions: blockchain can significantly assist in this. The success of the interaction between the two technologies is anyone's guess. However, there is a strong and rapidly developing potential for real disruption.

6. Challenges of merging AI and blockchain

AI solutions vary from legacy solutions because they adopt probabilistic models. The standard program, in other words, follows "If A happens, THEN follow B." AI, in comparison, uses probabilistic responses to make a good move. This feature makes AI the perfect technology for versatile solutions. The concession, however, is that such AI systems make errors (Pinto, 2018).

To date, AI agents often appear to go wrong, and it remains difficult for users to know whether they are mistaken. Some unforgettable examples include a rogue Microsoft chatbot, Wikipedia edit bots feuding with each other, Uber's self-driving cars ignoring red lights, and Promobot IR77, the Russian robot that escaped from the laboratory (Pinto, 2018).

Compliance is another concern (Pinto, 2018). Regulation of AI solutions to prevent causing harm continues to be a significant concern. AI and blockchain solutions need a real challenge for data aggregation. The IoT is nevertheless essential in providing the data necessary for AI training. In reality, the protection and privacy of data in this field will be crucial.

Talent is another obstacle to face when connecting blockchains and AI (Pinto, 2018). While IoT devices can collect data, training AI models is secondary, and it is necessary for AI models to run in a decentralized or distributed form. Fortunately, like Deep Brain Chain and SingularityNET, other businesses are still looking for ways to improve their algorithms.

Computing resources remain a challenge when AI and blockchain are combined. Fortunately, it is possible to utilize global idle computing resources to conduct resource-intensive blockchain AI training.

7. The integration of AI and blockchain for industry 4.0

Over time, blockchain and AI have formed their pathways with a slight overlap. There is a direct, coordinated link between the two technologies in the form of data. However, according to a survey, the future of banks and insurance companies is primarily ordered to shift from blockchain to distributed AI. The loss of significant consumers would then be just one of the possible disruptive situations, and industry 4.0 would be affected by the same situation.

In comparison, AI has sliced through the embryonic stage of advanced technology development, a distinct but related path. Its usage and acceptance have increased, and its implementation has reached a critical stage of organization. Interfacing one or more blockchains to banks and insurance undertakings will speed transactions, improve protection and accountability, and optimize data control (Kumar, 2019).

1. ***Integrating AI and blockchain in rising performance:*** AI and blockchain can gain and endorse each other to move into the fourth industrial revolution. For example, a chatbot can handle customer service if a client wants clarification on a specific subject or has a particular request.

As ML and algorithms push chatbots, the solution is certainly more rewarding and quicker. Here, the AI approach will evolve, and a channel is needed to achieve this purpose, where those customers who have a direct interest can share information and communicate in complete protection.

2. ***Integration offers a strategic advantage for businesses:*** The reason behind the AI revolution is big data progress. It helps organizations organize a massive dataset of organized components that can be processed very quickly by computers. Simultaneously, this data value has encouraged blockchain development because its distributed directory is a revolutionary way for data to be stored alternatively and effectively. The integration of AI and blockchain impacts various areas, including the following.
 - a. *Security:* AI and blockchain can provide a double shield against cyberattacks. Finding suspicious incidents is one of the most significant obstacles for prospective companies.
 - b. *Speed:* Using both technologies will increase information processing and data transfer speed and allow consumer experiences with companies to be more comfortable and quicker.
 - c. *Usability:* Services can be more personalized in the coming years and be the framework for large companies' suggestions.
3. ***What incorporation implies for industry 4.0:*** It has been seen that blockchain solves a variety of inefficiencies in existing technology and AI efficiency due to the value of data currently available. The prevalent delivery of AI and the introduction of blockchain will also be significant for industry 4.0. At present, companies address software, algorithms, automation, robotics, and hardware and discuss more complicated topics such as on-demand design and development of goods, dematerialization, and disintermediation. Industry 4.0 will be the first big transition to switch from one technologically centered state to another, possibly being more advanced, emphasizing various cornerstones like information transparency, assistance, and interconnection.

8. How the app development industry is using blockchain and AI integration to innovate

The application development industry combines AI and blockchain into its newly developed applications. The advantages can be seen in the following points (Laura, 2020):

1. **Encrypted security:** The data held in the blockchain would be protected with inherent encryption. If anyone wants to encrypt highly sensitive information, like their bank accounts and other confidential records, blockchain is what any developer should think of as the best technology. In the meantime, using AI would be easier to find documents with a single click, because it collects a high volume of data. Researchers worldwide are now working on ways of using AI to prevent security from being encrypted.
2. **Unengaging the AI thinking process:** The user will only benefit from AI if he knows using and understands machine language. However, with blockchain, every change and every click is registered, making it easier to use the software or system.
3. **Managing and accessing data market:** Market players such as Google and Facebook have access to a lot of data, which will be very useful in AI processing. However, smaller industries continue to search for AI. When designing applications, incorporating problems with the blockchain would help record all changes and data entered and facilitate users in imitating the behavior using AI.
4. **Energy optimization and smart contracts:** For energy use optimization, AI requires training on historical data from data centers, and technology must be incorporated into organizations. The same idea also applies to the mining industry and would further slash mining hardware costs. Any glitches must be treated carefully, and immediate action should be taken.

Integration of these two technologies supports both the Android and iOS app developers. For example, blockchain's incorporation has turned the payment portal into a smooth channel, and AI defines payments. Security is a serious problem that has hit mobile applications but can be solved using blockchain and AI.

9. Blockchain and AI building a powerful match

Various companies merge blockchain and AI with excellent results (Daley, 2019; Choudhury, 2020).

AI and blockchain are two of the most popular and elegant techniques. When combined, the two common techniques can deliver robust results and can be used for several purposes, including supply chain management, financial security, and various data sets. Following are the top projects where companies have merged blockchain and AI.

1. **FINALIZE:** The software platform Finalize creates applications to enhance civil infrastructure using blockchain and ML. It automates and speeds up workflow, verification procedures, and the construction industry's management, integrating with wearables to satisfy safety regulations.
2. **BLACKBOX AI:** Blackbox AI designs new AI tools. The organization's engineers build a custom information infrastructure that facilitates computer training and language processing for blockchain tools. Further building blockchain technology, the company also provides consulting services for its products to unlock blockchain potential.
3. **CORE SCIENTIFIC:** Core Scientific combines customized AI and blockchain infrastructure among existing business networks to update the business's servers, infrastructure, and apps that get real-time data reporting from supply chain management.
4. **BEXT360:** Bext360 uses AI and blockchain to increase accountability and productivity in the coffee, wood, seafood, and mineral industries' supply chains. AI analyses crops and forecasts growing trends, whereas blockchain ensures that from the seed to the finished product, the supply chain is documented.
5. **BURSTIQ:** BurstIQ developed a "Health Wallet" incorporating blockchain, AI, and big data for comprehensive patient data handling. BurstIQ wallet provides a patient healthcare team entry to their insurance history and health plans. Health professionals may then purchase, exchange, or sell patient data for various clinical experiments or to learn about a particular disease. Conversely, the blockchain helps patients keep personal identity details secret, even exchanging only large-scale health data.
6. **VIA:** VIA integrates blockchain and AI to enable some of the biggest potential firms worldwide to design and use data more efficiently. The Trusted Analytics Chain secures data of a power utility, secretly collects from various locations and firms, and uses smart contracts to give businesses more profound insights into the potential market as a whole.
7. **COINGENIUS:** CoinGenius is an AI-driven cryptocurrency trading platform that offers AI-based scoring, high-quality data feeds, and advanced projections for cryptocurrency traders to increase their potential by tailoring news and trading features.
8. **ALPHANETWORKS:** A new entertainment platform that utilizes AI and blockchain to increase user insights and viewership algorithms is AlphaNetworks. This company's proof-of-engagement tools use AI and blockchain to enable automatic payments to content creators.

9. **NETOBJEX:** NetObjex is a smart urban technology network using AI, IoT, and blockchain to power connected devices and cloud-based goods. These technologies are allegedly combined to enhance logistics monitoring, data and system authentication, and real-time failure detection.
10. **FIGURE:** Figure incorporates blockchain and AI into the home loan system to accommodate new entry points for consumer credit products. These solutions include home equity lines of credit, home remodeling loans, and even home buy-lease offers. According to the organization, loans can be made available to customers as soon as they have filled out a loan application.
11. **WEALTHBLOCK.AI:** Wealthblock.AI is a SaaS (software as a service) platform that enables organizations seeking financing to handle automated marketing and messaging. Firms will monitor and operate their suitability checking method using a blockchain to ensure their business partners are of excellent quality. More than that, Wealthblock's AI assists with content creation, keeping investors in the loop at all times.
12. **CYWARE LABS:** Cyware Labs integrates AI and blockchain-based threat intelligence solutions. One example of supporting mobile threat intelligence, safe messaging, and suspicious incident reporting is the Cyware Situational Awareness Platform. With this approach, users may inspect their current real-time mobile security protocols in great detail.
13. **CHAINHAUS:** Blockchain and AI consultancy education and marketing firm Chainhaus offers everything from training and software expansion to research, and capital raising is covered in a full spectrum of services provided by the firm.
14. **VERISART:** Using AI and blockchain, Verisart can identify and authenticate fine art in real time. Artists can establish tamper-proof certificates of work that ensure credit and immutability using the company's blockchain. By using blockchain technology and time-stamping, these certificates do not allow fakes to be sold as originals.
15. **BOXSPRING MEDIA:** Boxspring Media claims to generate original programming for new-starting careers, which is interactive, on-demand, and is made to be viewed while employed. To help employees stay current on industry developments, each Boxspring Media platform features a specific subject (such as networking, new technology, etc.) to supply employees with relevant and consumable material.

16. **VYTALYX:** To give healthcare professionals blockchain-based access to medical knowledge and insights, Vytalyx, a health technology company, uses AI. By storing medical information on a blockchain, a patient's medical staff may instantly use AI to search for clinically relevant medical studies, personalize medications and diet plans, and enhance patient and doctor contact.
17. **WORKDONE:** Workdone automates processes between SaaS platforms. The "work heuristics management" company uses ML and smart contracts to gather work-related, customer-specific features from entire SaaS platforms in a business ecosystem to lessen the time spent on repetitive tasks such as customer retention and analysis processes.
18. **STOWK:** A blockchain-based platform by Stowk aims to speed up data access, IT governance, and procurement by using AI.
19. **MOBS:** Mobs is a blockchain-based video marketplace where mobile videos are sold and purchased. When a consumer sells his video, potential customers—claim advertisers who need to have convincing footage—can click through the application to find their needs. Mob's blockchain then produces a smart contract that allocates money directly to the producer of content based on interaction rates and views.
20. **HANNAH SYSTEMS:** Hannah Systems is creating AI and blockchain technology for self-driving automobiles. A car equipped with ML may rapidly access, examine, and securely store data using the company's data-sharing network, an analytics dashboard, real-time mapping tool, and a blockchain.
21. **LIVEEDU:** LiveEdu is an online learning platform where students learn how to create authentic goods by getting training with the developers who make them. The platform offers classes on all elements, including the implementation of AI and the fabrication of Bitcoin development instructions. It delivers on what it promises as LiveEdu incorporates blockchain and smart contracts to benefit authors of the content.
22. **OBEN:** ObEN built a technology that enables intelligent virtual avatars in Augmented reality (AR) and virtual reality (VR). Anyone can build their virtual avatars for communication on the platform. To provide a more personal and social experience, AI, blockchain, and VR all work together.

23. **NEUREAL:** Neureal is a prediction engine that incorporates AI, blockchain, and cloud computing to forecast various things, including stock market movements and Google search data. In predicting past events, the company's AI scans past projections to anticipate possible future events. In addition, the blockchain database keeps account of every outcome, and the computer networks may search for patterns to see if their predictions are correct.
24. **NUMERAI:** This is a decentralized hedge fund where over a hundred data scientists worldwide collaborate on AI and blockchain matters. As part of the ongoing NUMERAI ML competition, Ethereum and ML data are made available to the researchers, creating ML models based on the data. Only the most sophisticated scientists receive Numeraire (NMR) crypto.
25. **COMPUTABLE:** Computable Labs dedicates itself to building blockchain-based solutions for AI marketplaces. By facilitating the sharing of data and algorithms on the decentralized AI marketplaces, the company empowers individuals to have more access to a larger range of information.
26. **SYNAPSE AI:** The Synapse AI network gives participants tokens (SYN Tokens) to exchange data when users share it. Users who use the Synapse Marketplace to exchange data may also acquire Bitcoin, Ethereum, or US dollars in return for their SYN Tokens. By leveraging the ability to share data as part of the reward system, Synapse AI's data-sharing marketplace provides users with the opportunity to contribute their data to a larger pool of available data. This makes data purchasers in the advertisement, education, entertainment, and pharmaceutical industries better able to access a more comprehensive data supply.
27. **DOPAMINE:** The market for AI and data providers' intellectual property is driven by Dopamine. The company's blockchain records each data provider's reputation and allocates incentives based on this information. By sharing insights, databases, and AI and blockchain technology, data providers can make money by solving problems and becoming important collaborators to the Dopamine community.
28. **BOTCHAIN:** BotChain facilitates uniform software registration, assessment, audit, and collaboration of AI agents. AI-enabled bots are certified on the blockchain BotChain ecosystem to log all data and get "BotCoin" by running apps on the distributed network.

29. **AI BLOCKCHAIN:** The AI blockchain uses intelligent digital agents to maintain the network. The blockchain company's platform and ledger are used to maintain the security of all the major businesses, from real estate to media to healthcare and the financial sector. Next, the firm uses AI to handle everything from developing and administering blockchains to expanding and maintaining the network.
30. **BLACKBIRD.AI:** Blackbird.AI, with its usage of AI and blockchain, rates the authenticity of news content. When confronted with "fake news," the organization relies on AI to analyze content and identify potentially false, hateful, or satirical content. Blackbird.AI blockchain acts as a database where the verified results of AI and crowdsourced reports are recorded.
31. **GAINFY:** The Gainfy healthcare platform combines AI, IoT devices, and blockchain technology to improve the healthcare business experience. As with other industries, there are various ways that this business incorporates new healthcare technology developments, including a digital urgent care network, an identity verification system, a data encryption tool, a crypto payment system, and a clinical trials database.
32. **AICOIN:** AICoin is a financial term that combines AI modeling advantages with a token. In this project, developers have developed AI models to define trading trends in about 20 liquid cryptocurrency markets. According to the developers, AICoin aims at enabling investors to create wealth through AI and blockchain.
33. **DEEPBRAIN CHAIN:** A decentralized, low-cost AI computer network that uses blockchain technology and privacy-preserving technology is called DeepBrain Chain. It is a decentralized neural network in essence. The purpose of this network is to provide a decentralized AI development cloud network. It is also a value-maximizing data network that provides data ownership and consumption as separate entities. NEP-5 chains will be replaced with a native substratum token with decentralized governance.
34. **MATRIX AI:** AI-based technologies such as natural language processing are used to fulfill the blockchain pledge. Matrix AI utilizes autonomous AI contracts, state-of-the-art cybersecurity, dynamic blockchains, and more. A protected virtual machine is driven by AI that commonly finds prospective vulnerabilities and hostile intentions, helping to keep the system robust during a large-scale attack.
35. **NAMAHE:** Namahe is a blockchain-based supply chain solution. It combines AI with blockchain technology to create a safe ecosystem

that saves on expensive audits and increases supply chain performance. Namahe improves supply chain productivity by allowing AI layers to seamlessly track the supply chain, seeking trend anomalies for detecting fraud, delays, irregular events, and flagging the analysis data.

36. **PECULIUM:** Peculium is an entirely open, decentralized AI and ML savings management platform. It seeks to help its users overcome the risks of the current investment market. Peculium also provides users with cryptocurrency asset tracking and control. The goal of the project is to optimize profits and savings based on ML. This is done by the financial advisor AIEVE of Peculium, i.e., AI, ethics, principles, and balance.
37. **SINGULARITY NET:** The whole-stack AI solution Singularity-NET is a decentralized protocol-driven solution. It is the only decentralized forum for AI to collaborate and organize on a scale. The SingularityNET network is the central framework for interacting and transacting AI services. SingularityNET helps everybody to benefit from an international network of AI algorithms, resources, and agents.

10. Conclusion

The emergence of blockchain and integrating AI into conventional services and products has created an entirely new range of possibilities for any of these powers to join hands and collaborate. The convergence of AI and blockchain produces possibly the world's most secure technology-driven decision-making systems that are practically tamper-proof and offer reliable insights and decisions. Over the years, businesses have spent heavily on AI-driven automation software, and today, with the advent of blockchain, it is possible to trust the interaction of machines with the system. Soon, we will witness the widespread acceptance of stable and intelligent business and governance environments aided by AI-integrated blockchain networks.

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